



Reality Check & Dataset Analysis

The Professor's Question: Can we combine Dataset 1 (Wellbeing & Lifestyle) and Dataset 2 (Health Activity)? **The Brutally Honest Answer:** No. Attempting to combine these datasets is statistically invalid, logically flawed, and structurally impossible.

Here is exactly why, and what you must tell your professor to defend your architectural decision.

1. Why Combining Them is a Catastrophic Error

There are only two ways to combine datasets in Data Science: Horizontally (Merging/Joining) or Vertically (Stacking/Concatenating). Both will fail here.

Reason A: Horizontal Merge Failure (Row-by-Row)

To merge two datasets row-by-row, you need a **Primary Key**—a unique identifier present in both datasets (like a Patient ID, Email, or SSN).

- **Fact:** Dataset 1 is an anonymous Kaggle survey. Dataset 2 is a separate anonymous dataset with its own sequential ID (1, 2, 3...).
- **Conclusion:** You cannot prove that Row 1 in Dataset 1 is the same human being as Row 1 in Dataset 2. If you merge them blindly, you are taking the stress score of a 40-year-old woman in Dataset 1 and randomly attaching it to a 56-year-old man's blood pressure from Dataset 2. **You will create Frankenstein patients that do not exist in reality.**

Reason B: Vertical Stack Failure (Column Schema Mismatch)

To stack datasets on top of each other, the columns must represent the exact same data types and scales.

- **Fact:** Let's look at how they measure the same concepts:
 - **Age:** Dataset 1 is Categorical ("36 to 50"). Dataset 2 is Continuous Integer (56).
 - **BMI:** Dataset 1 is Ordinal (1 or 2). Dataset 2 is Continuous Float (30.72).
 - **Steps:** Dataset 1 is Binned (e.g., 5 meaning 5,000). Dataset 2 is Exact (5134).
- **Conclusion:** If you stack them, your algorithms will crash because the data types clash, and the variances are completely incompatible.

2. Decoding Dataset 2: The Saving Grace

While you cannot combine them, **Dataset 2 is the exact dataset you need to fulfill your project proposal.** Your original proposal stated your objective was to: *"Predict the type of disease that person might get with a precision score."* Dataset 1 had no diseases. Dataset 2 has them.

Data Dictionary (Dataset 2 Highlights)

- **Continuous / Raw Inputs:** Age (Int), Height_cm (Int), Weight_kg (Int), BMI (Float), Daily_Steps (Int), Calories_Intake (Int), Hours_of_Sleep (Float), Heart_Rate (Int), Exercise_Hours_per_Week (Float).

- *Why this is great:* You have exact variance. You don't have to deal with annoying categorized bins like in Dataset 1.
- **Categorical Inputs:** `Gender` (Male/Female), `Smoker` (Yes/No).
- **The Problem Child (Requires Engineering):** `Blood_Pressure` (e.g., "137/72").
 - *The Issue:* This is a string (object). A machine learning model cannot multiply or split a string.
- **The Target Variables (Clinical Labels!):** `Diabetic` (Yes/No), `Heart_Disease` (Yes/No).

3. The Strategic Pivot (My Recommendation)

Tell your professor this:

"After analyzing both datasets, combining them is statistically invalid because they lack a common primary key, and their feature scales (continuous vs. binned) are incompatible. Instead, I propose a Dual-Model Architecture."

1. **Use Dataset 1 (Wellbeing) strictly for Phase 4 & 5 (EDA and Correlation Analysis)** to explore psychological/lifestyle correlations (Stress, Flow, Social Network).
2. **Use Dataset 2 (Health Activity) for Phase 8 (Machine Learning).** Because Dataset 2 contains actual binary disease labels (`Diabetic` , `Heart_Disease`), we can finally switch to a **Classification Pipeline** (Logistic Regression, Random Forest Classifier) to actually predict disease risk, perfectly aligning with our research proposal.

4. ASSESSMENTS & TASKS

Dataset 2 requires a completely different preprocessing technique because of the `Blood_Pressure` column.

Mini Assignment 8 (Data Engineering): In medicine, Blood Pressure consists of two separate numbers: Systolic (the top number, e.g., 137) and Diastolic (the bottom number, e.g., 72).

1. Write a Python script using pandas to split the `Blood_Pressure` string column into **two brand new numerical columns:** `BP_Systolic` and `BP_Diastolic` .
2. Drop the original `Blood_Pressure` string column.
3. Convert the `Diabetic` and `Heart_Disease` columns from "Yes"/"No" strings into `1 / 0` integers.

Interview Question: If you build a Classification model using Dataset 2 to predict `Diabetic` (Yes=1, No=0), and the dataset has 900 people without diabetes and only 100 people with diabetes, your data is "highly imbalanced." If your model simply guesses "No" for every single patient, what will its Accuracy be? Why is Accuracy a terrible metric in this scenario, and what metric should you use instead?

Provide your Python code for the `Blood_Pressure` split and your answer to the interview question.